MSEC 2025 | NAMRC 53

June 23-27, 2025, Greenville, South Carolina

CLEMSON UNIVERSITY

Arizona State University

CRONET: A CONVOLUTIONAL RECURRENT OPERATOR APPROXIMATOR NETWORK TO ACCELERATE TOPOLOGY OPTIMIZATION

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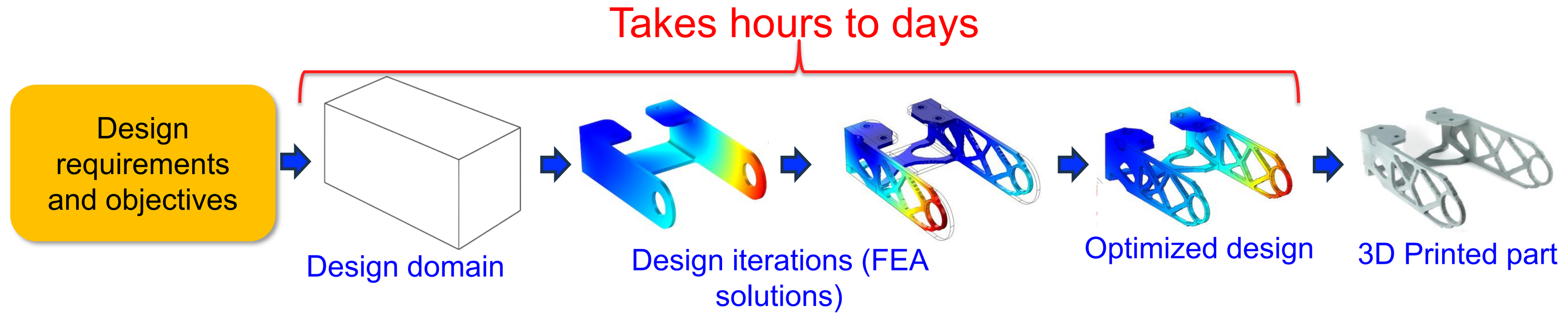
ACCELERATING TOPOLOGY OPTIMIZATION (TO)

- Design optimization is critical across industries (e.g., aerospace, energy, health)
  - Used to create parts that have improved efficiency, reliability, performance and satisfaction
  - Traditional processes take months to years to optimize a design
- **TO** offers significant potential to accelerate this lengthy design cycle.
- It auto-generates ultralight, high-performance geometries

Takes hours to days



- However, classic TO algorithms requires hundreds of time-consuming FEA solve per job creating major design bottleneck

How can we accelerate TO and eliminate this bottleneck?

TO PROBLEM FORMULATION

- Classical SIMP formulation [Bendsoe, 1989]

$$\min_{\mathbf{x}} c(\mathbf{x}) = \mathbf{U}^{\mathrm{T}}\mathbf{K}\mathbf{U}$$

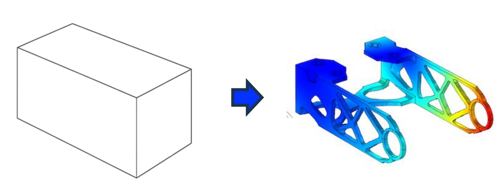
$$\text{s.t.} \quad \frac{V(\mathbf{x})}{V_0} = f, \qquad \mathbf{K}\mathbf{U} = \mathbf{F}, \qquad x_{\min} \le x_e \le 1$$

Compliance; Global stiffness matrix; Global displacement matrix; Load matrix

OBJECTIVE

- Distribute material inside Ω to minimize compliance under the loading conditions subject to the given constraints

TO finds optimal design



TRADITIONAL SOLUTION APPROACHES

Algorithm	Key Features
Sequential Linear Programming (SLP) [Gomes & Senne, 2011]	linearize objective / constraints; good for small N, stalls on large meshes
Method of Moving Asymptotes (MMA) [Svanberg, 1987]	handles many constraints; heavier per-iteration cost
Optimality Criteria (OC) [Bendsøe, 1995]	simple closed-form update, faster on high-resolution grids

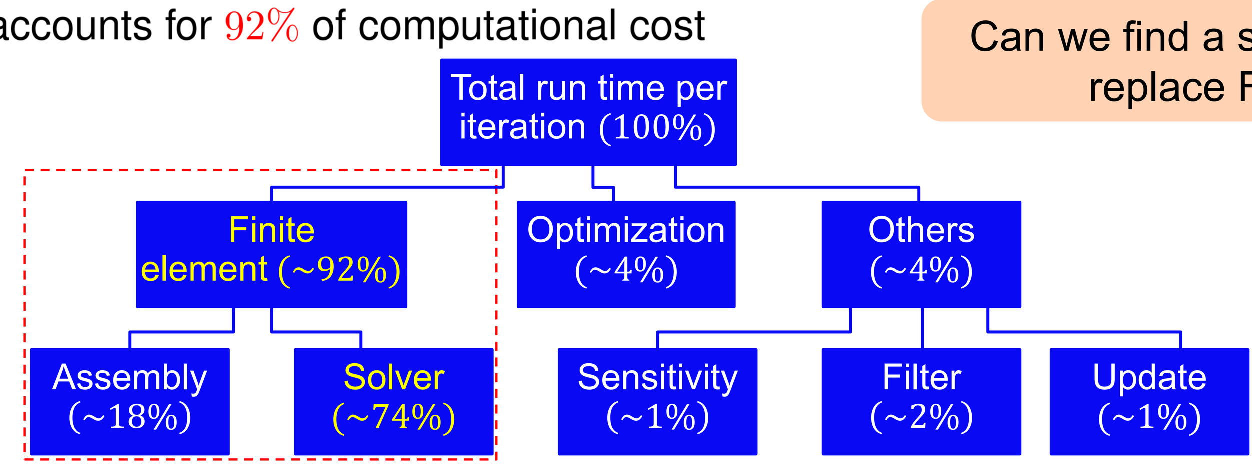
- OC method updates design densities via simple local scaling rules
- Move limits and damping enables stability and convergence.

Challenges
- Takes long time to optimize design even for small problem
- Computational costs explode non-linearly with number of elements

COMPUTATIONAL BOTTLENECK

- FEA accounts for 92% of computational cost

Can we find a surrogate to replace FEA?



DEEP OPERATOR NETWORK

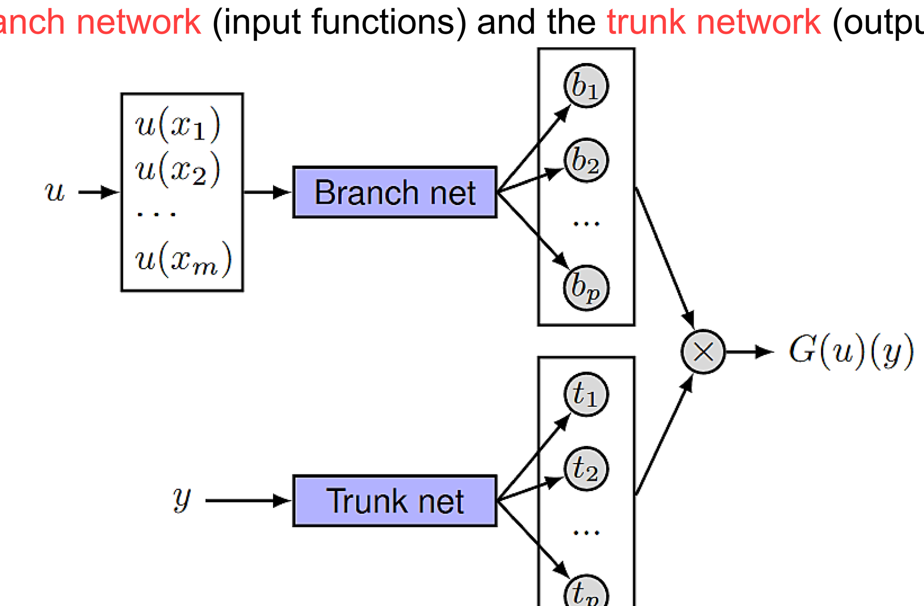
$$\left| G(u)(y) - \underbrace{\sum_{k=1}^{p}\sum_{i=1}^{n} c_i^k \,\sigma\left(\sum_{j=1}^{m} \xi_{ij}^k u(x_j) + \theta_i^k\right)}_{\text{branch}} \underbrace{\sigma(w_k \cdot y + \zeta_k)}_{\text{trunk}} \right| < \epsilon$$

Nonlinear continuous operator; Output location; Input function; Neural approximations

ABSTRACT:

Emerging additive and hybrid manufacturing, coupled with open CAD/CAM datasets, underpins social manufacturing, where users adapt and fabricate designs on demand. Topology optimization supports this vision by allocating material for performance objectives but depends on computationally expensive finite-element analysis (FEA), slowing rapid prototyping of complex parts. In this work, we replace FEA with CRONet—a Convolutional Recurrent Operator Approximator Network that learns 2-D/3-D operator mappings directly, marrying convolutional layers for spatial correlations with recurrent layers that follow the sequential design updates. CRONet delivers high-fidelity compliance and thermal estimates while slashing computational cost. In benchmark tests on the Messerschmitt-Bölkow-Blohm and cantilever beams, CRONet cuts runtime by up to 78% yet keeps compliance within 5% of traditional FEA. This surrogate makes high-resolution, multi-physics topology optimization practical for enthusiasts, SMEs, and large-scale additive manufacturing workflows, accelerating the feedback loop central to social manufacturing.

- A neural network for operators based on universal approximation theorems.
- DeepONet composed of two sub-networks) [Lu et al., 2019]
  - The branch network (input functions) and the trunk network (output locations)

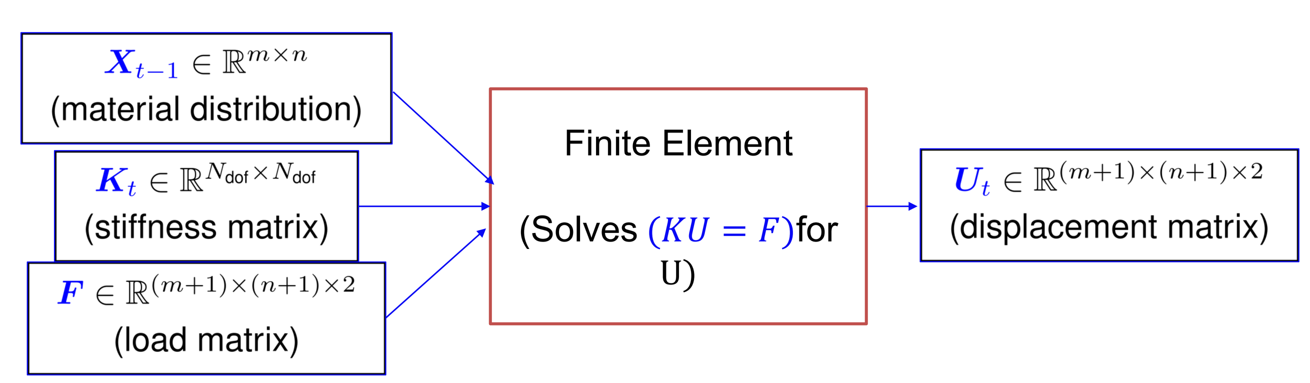


DeepONet Architecture

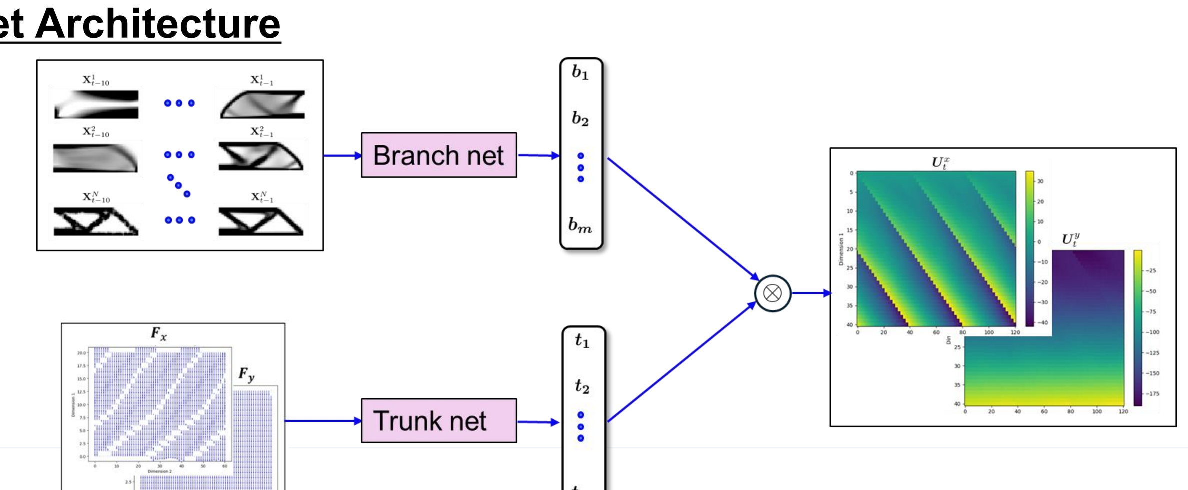
- Limitations: can not handle the 2D/3D requirements of topology optimization

PROPOSED METHOD: CRONET, 2D/3D OPERATOR NETWORK

- Finite element structure



CRONet Architecture



- CRNN in branch leverages **time series** data of material densities
- CNN layers in trunk encodes load field information

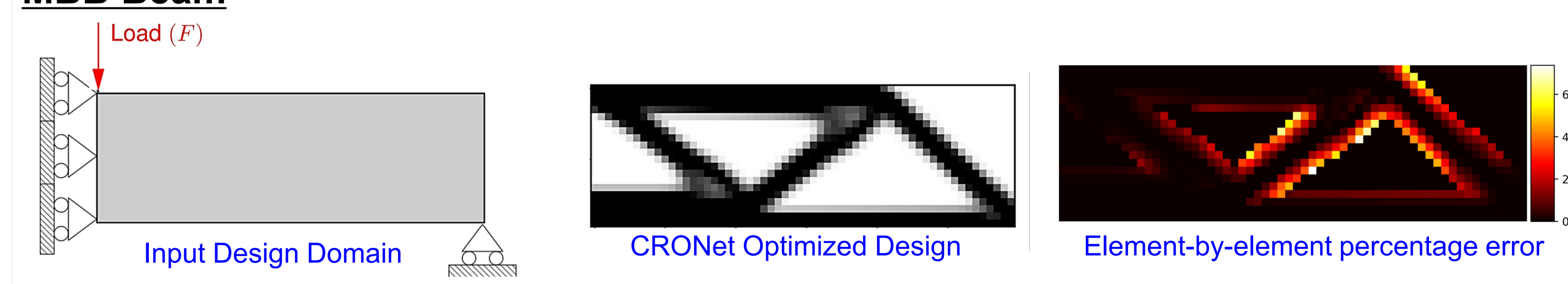
CRONet Integration

- First 20 FEA iteration creates training data
- Adopt Structural similarity index (SSIM) loss
- Trained CRONet replaces FEA in the loop
- FEA corrects prediction when decay occurs

CRONet creates rapid convergence with FEA-level accuracy.

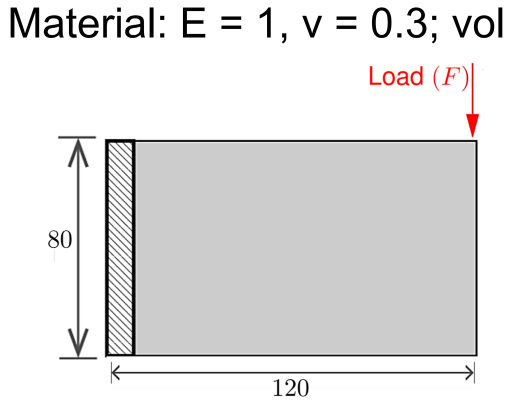
CASE STUDY

MBB Beam

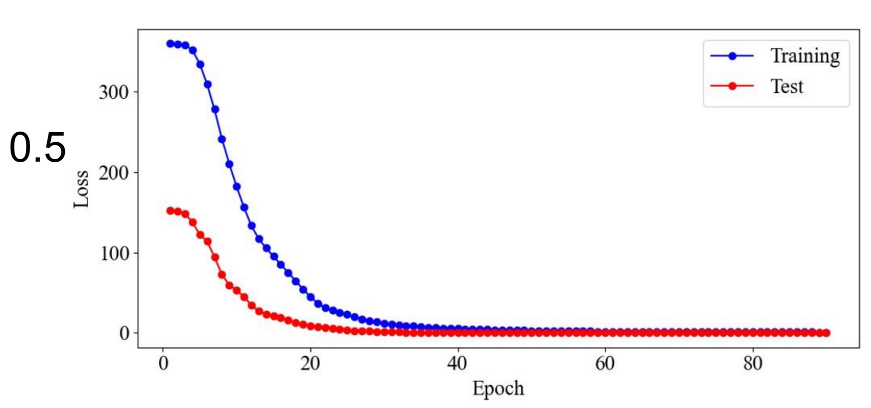


Cantilever Beam

- Rectangular 120 × 80 mesh,
- Left edge fixed, Downward tip load
- Material: E = 1, ν = 0.3; volume frac = 0.5

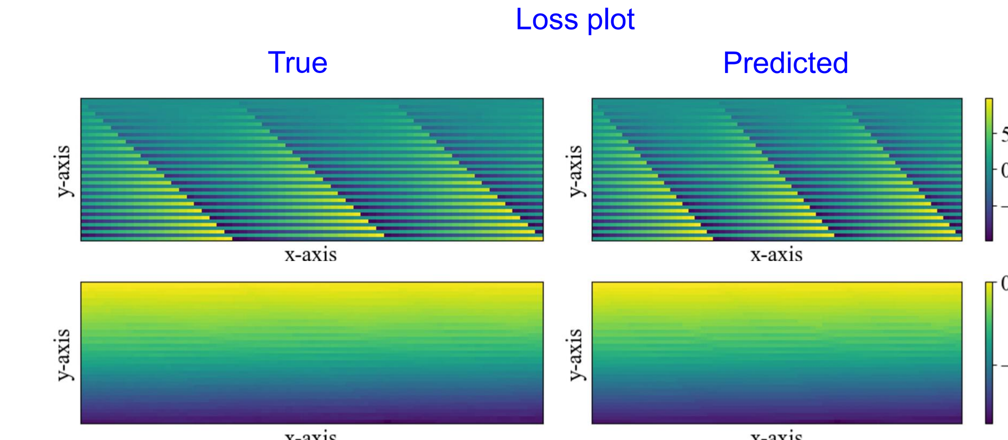


Cantilever beam design domain



Loss plot

CRONet optimized design

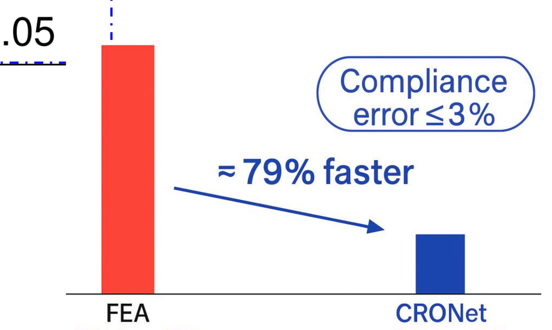


Displacement comparison

PERFORMANCE SUMMARY

Metric	MBB		Cantilever	
	FEA	CRONet	FEA	CRONet
Total optimization time (s)	356	81	1792	370
Average iteration time (s)	1.20	0.26	5.80	0.94
Time reduction (%)	—	78	—	79
Iterations to convergence	297	317	304	328
Training time (s)	—	150	—	760
Convergence error	—	0.03	—	0.05

- 79 % run-time reduction
- Compliance ≈ 3 %
- Optimized design within 5% of FEA



CONCLUSION

- CRONet accelerates topology optimization
- It provides **≈ 80 % reduction** in computation time
- Optimized design compliance error **≤ 3 %**

FUTURE WORK

- Improve generalization capabilities with minimal training samples and extend CRONet to multiple load cases, dimensions
- We will also apply the method to multi-physics / multi-objective topology optimization

References

- Bendsoe, 1989, Structural optimization 1, 193–202 (1989).
- Lu, Lu, Pengzhan Jin, and George Em Karniadakis. arXiv:1910.03193 (2019).